

Straight Lines

Week 4A

Overview

Introduction and Graphing Straight Lines

Key Ideas - Introduction to Straight Lines

- Graphing Straight Lines

Key Idea – Introduction to Straight Lines

Probably the most important equation and graph that you will use are those associated with the **straight line**. A large number of relationships between engineering variables can be described using a straight line or linear graph.

Every function of the form $y = f(x) = mx + b$, where m and b are constants is called a **linear function**. The constant m is called the **coefficient of x** , and b is referred to as the **constant term**.

Examples:

$$f(x) = 2x + 1, \quad g(x) = -\frac{1}{2}x + 3 \quad \text{and} \quad h(x) = -3x \quad \text{are all linear functions.}$$

The graph of the linear function is a straight line.

Graphing a Linear Function by plotting Points

On a number plane, the **graph of a linear equation** is a **straight line**. We can graph a linear equation by constructing a 'table of values' and plotting the points on a number plane. For a straight line, a **minimum of two points** is required. However it is safer to use more points. For simplicity we often choose $x = 0, 1, 2$.

To **draw the graph of a linear equation**, follow these steps:

Step 1 Select values of x . Choose $x = 0, 1, 2$.

Step 2 Create a table of values by substituting the chosen x values into the equation to calculate the corresponding y -value.

Step 3 Summarise the values from steps 1 and 2 as points with coordinates (x, y) .

Step 4 Plot the (x, y) points. Draw a straight line through these points.

Step 5 Ensure that the line and axes are correctly labelled.

Example: Sketch the line with equation $y = x + 2$ by preparing a ‘table of values’.

Step 1 Select $x = 0, 1, 2$.

Step 2 Create a table of values by substituting $x = 0, x = 1$ and $x = 2$ into the equation $y = x + 2$ to find the corresponding y -values.

If $x = 0$ then $y = x + 2$ is $y = 0 + 2 = 2$

If $x = 1$ then $y = x + 2$ is $y = 1 + 2 = 3$

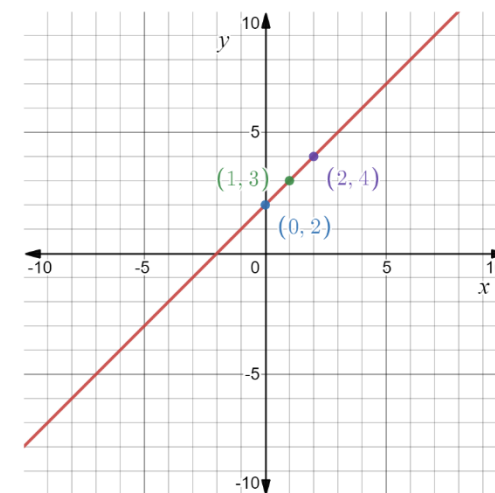
If $x = 2$ then $y = x + 2$ is $y = 2 + 2 = 4$

x	0	1	2
y	2	3	4

Step 3 Summarise the values from steps 1 and 2 as coordinates ie $(0, 2)$, $(1, 3)$ and $(2, 4)$.

Step 4 Plot the (x, y) points i.e. $(0, 2)$, $(1, 3)$ and $(2, 4)$ and draw a straight line through these points.

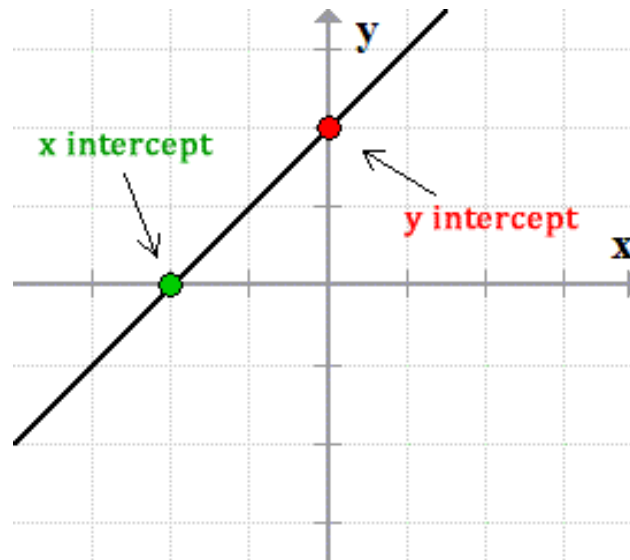
Step 5 Label the line and axes.



Note: This method is referred to as “preparing a table of values.”

Graphing a Linear Functions using Intercept Method

A more common method is to use when graphing linear functions is to determine **the x -intercept and the y -intercept**, then **joining these points to sketch the line**.



Note:

The x -intercept is where the line cuts the x -axis and the y -intercept is where the line cuts the y -axis.

Discussion Question: How could you find the coordinates of the x and y intercepts given an equation such as $f(x) = 2x - 1$?

- for x -intercept: let $y = 0 \rightarrow 0 = 2x - 1 \rightarrow x = \frac{1}{2}$
- for y -intercept let $x = 0 \rightarrow y = 2 \times 0 - 1 \rightarrow y = -1$

From the above we can conclude, the intercept method consists of finding where the line cuts the x and y axes, then joining these points to sketch the line.

When using the **intercept method** follow these steps:

- Step 1** Find the x -intercept by setting $y = 0$ and solve for x
- Step 2** Find the y -intercept by setting $x = 0$ and solve for y
- Step 3** Draw a line through these two points

Example:

Draw the line $y = \frac{1}{2}x + 2$ by finding the x and y intercept.

x -intercept: set $y = 0$ and solve for x :

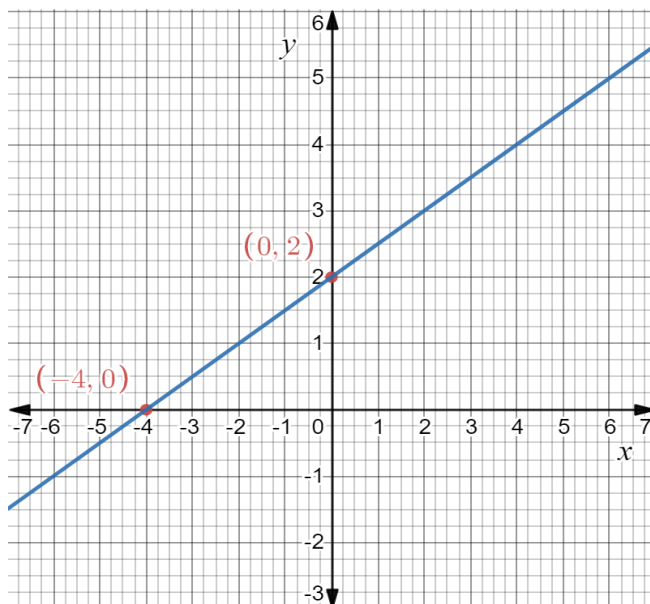
$$\frac{1}{2}x + 2 = 0$$

$$x = -4 \quad \therefore (-4, 0) \text{ is the coordinate of the } x\text{-intercept}$$

y -intercept: set $x = 0$ and solve for y :

$$y = 0 + 2$$

$$y = 2 \quad \therefore (0, 2) \text{ is the coordinate of the } y\text{-intercept}$$



Exercise: Sketch the graph of the linear equation $f(x) = -\frac{1}{2}x + 1$ using the intercept method.

• x -intercept (let $y=0$)

$$0 = -\frac{1}{2}x + 1$$

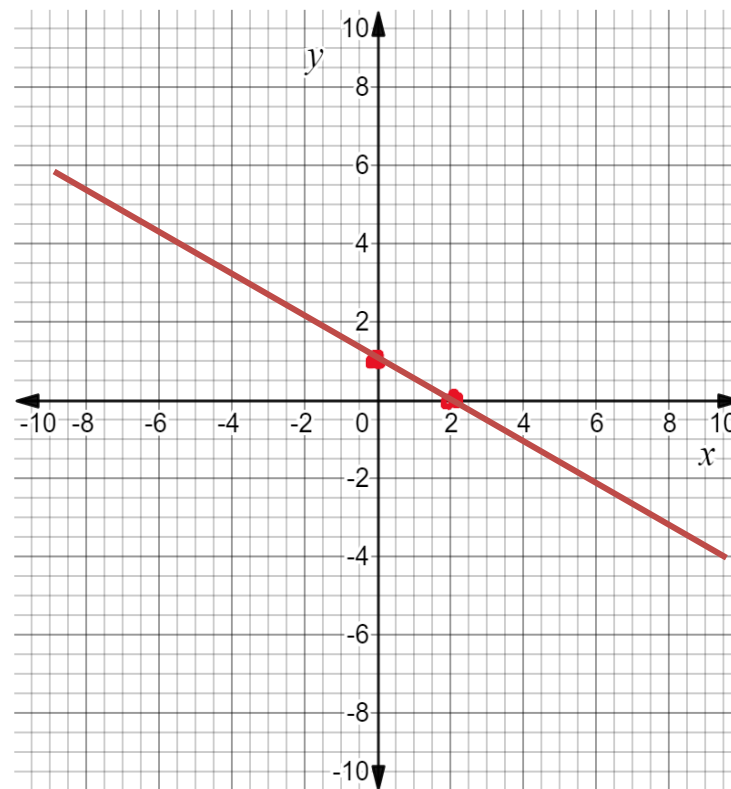
$$\frac{1}{2}x = 1$$

$$\boxed{x = 2}$$

• y -intercept (let $x=0$)

$$y = -\frac{1}{2} \times 0 + 1$$

$$\boxed{y = 1}$$



Condition for a Point to Lie on a Line

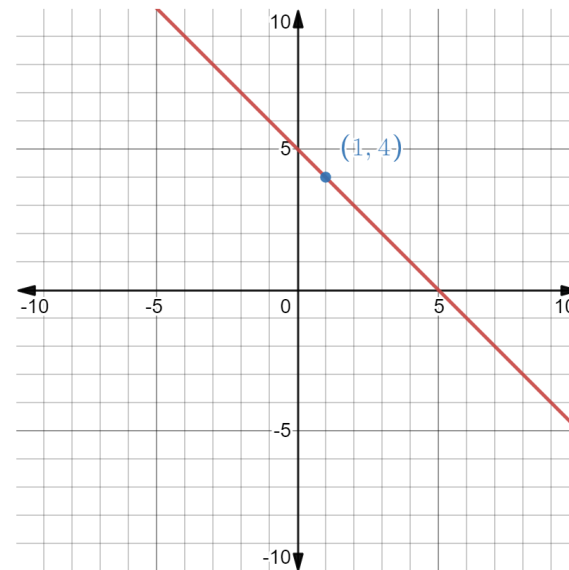
The coordinates of any point on a line **satisfy** the equation of that line. To determine if a point lies on a line, substitute the values of x and y into the LHS of the equation, then into the RHS of the equation, and finally check if the left hand side equals the right hand side.

Example:

Does the point $(1, 4)$ lie on the line $x + y - 5 = 0$?

$$\begin{aligned} LHS &= x + y - 5 \\ &= 1 + 4 - 5 \\ &= 0 \\ &= RHS \end{aligned}$$

Therefore, the point $(1, 4)$ lies on the line.



Exercise: Find w if $(w, -1)$ lies on the line $3x - 2y = 5$.

$$3 \times w - 2 \times (-1) = 5$$

$$3w + 2 = 5$$

$$3w = 3$$

$$\boxed{w = 1}$$